

Vivian Ha

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Profile

Software Engineering graduate with deep hands-on experience in applied machine learning, neural network design, and end-to-end model development. Current research focuses on deep learning and generative models for physical layer optimisation in wireless systems, with demonstrated ability to design, train, and deploy models from simulation to embedded hardware. Strong foundation in software engineering fundamentals, supported by industry R&D experience in translating technical research into commercial solutions.

Education

Master of Philosophy (Engineering) | *The University of Sydney*

2025 – expected June 2027

- Thesis: “End-to-End Learning with CWGAN Channel Surrogate and Neuromorphic Receiver for Energy-Efficient Wireless Communications”
- Research in 5G/6G physical layer optimisation, neural receivers, and energy-efficient communication for IoT

Bachelor of Engineering First Class Honours (Software) | *The University of Sydney*

2021 – 2025

- Honours thesis on generative adversarial network-based channel for 5G end-to-end learned systems

Professional Experience

R&D Intern | *Audinate*

Dec 2023 – Feb 2024

- Designed and prototyped an automated cloud API workflow for Dante Director, a commercial IoT/AV device management platform used across live events, broadcast, and building infrastructure
- Applied GraphQL API for real-time device tracking, remote configuration, and diagnostics across networked devices
- Collaborated with cross-functional teams (product, testing, engineering) using Agile methodology to scope and deliver beta features

Engineering Research Internship | *The University of Sydney*

Dec 2023 – Feb 2024; Dec 2024 – Feb 2025

- Designed and trained Machine Learning models for wireless channel optimization, validated on a standard 3GPP channel for performance benchmarking

Web Developer Student Intern | *Caritech Software Malaysia*

Jul 2021 – Aug 2021

- Performed front-end development, SQL database maintenance, and systematic testing/debugging of web applications in production environments

Key Projects

5G End-to-End Wireless Communication System with CWGAN-based Channel

Python, TensorFlow, NVIDIA Sionna, Embedded hardware (NVIDIA Jetson, USRP, OpenAirInterface)

- Developed end-to-end deep learning system, replacing traditional signal processing blocks with neural networks
- Deployed complete 3GPP 5G stack using OpenAirInterface (OAI) on NVIDIA Jetson with USRP software-defined radios, configured and tested 5G gNB and UE components for hardware-in-the-loop testing

AWS Serverless Cloud Image Processing Application

Python Flask, AWS Cloud (Lambda, S3, EC2, RDS, CloudWatch)

- Designed automated event-driven serverless architecture, achieving 75% success under 500 req/sec load testing
- Implemented three-tier security model with auto-scaling, multi-AZ (availability zone) deployment, and automated monitoring and alerting via CloudWatch

IoT Smart Environmental Monitoring System

Raspberry Pi, TypeScript, Cloud API

- Developed IoT solution with real-time distributed sensor data collection and edge processing on Raspberry Pi
- Implemented cloud-based data pipeline with automated analysis and AI-driven response delivery under 5 seconds

Skills

- **Programming:** Python, Java, C, JavaScript/TypeScript
- **AI/Machine Learning:** TensorFlow, TensorRT, Neural Network Design, Deep Learning for signal processing
- **Cloud & DevOps:** AWS, Serverless Architecture, CI/CD, Git, Jira, Bitbucket
- **APIs & Integration:** GraphQL, REST APIs, Spring Boot, React, PostgreSQL, MySQL
- **Systems & Platforms:** Linux/Ubuntu, NVIDIA Jetson, Raspberry Pi, Docker